



# TEST PLAN WORKSHEET

## PROGRAM DEPUTY APPROVAL

PD Signoff: [Signature] Date: 3/27/2000  
Ops Reviewer Signoff: [Signature] Date: 4-2-00  
Expiration Date (max. 90 days from approval): \_\_\_\_\_  
Presentation Required? ☐ yes ☐ no

## COMPLETION INFORMATION

Completion Date: \_\_\_\_\_  
Crew Chief Signoff: \_\_\_\_\_  
Comments (partial completion, etc.): \_\_\_\_\_

**NOTE:** Information addressing the appropriate content of each of the following sections can be found in Section 2.0 of the Test Plan Instructions.

**Test Plan Title:** Evaluate focusing at end of Hall A line

**Author(s):** Jay Benesch

**Date Submitted:** 3/27/2000

**Revision Number:** Rev. 0, (3/27.00 1400)

### Brief Purpose of Test

Evaluate focusing strength of quads 1H01 through 1H03A.

### Anticipated Benefits

so we don't spend as much time screwing around to get hall A beam profile right

### Beam Conditions Required

Complete all of the following tables, entering a value or an X in the appropriate spaces:

#### Beam Type/Current (enter value)

| Beam Type                                      | Beam Current |
|------------------------------------------------|--------------|
| Beam Off                                       |              |
| Pulsed (std. current = 8 $\mu$ A) <sup>a</sup> | X            |
| CW                                             |              |

a. The standard current for pulsed beam operation is 8  $\mu$ A. If your test requires pulsed beam current >8  $\mu$ A, then specify the required current and provide a brief explanation next to the specified current.

#### Beam Energy (select one)

| Beam Off | 845 MeV<br>(1-pass) | 1.645 GeV<br>(2-pass) | 2.445 GeV<br>(3-pass) | 3.245 GeV<br>(4-pass) | 4.045 GeV<br>(5-pass) | Other<br>(specify) |
|----------|---------------------|-----------------------|-----------------------|-----------------------|-----------------------|--------------------|
|          |                     | X                     |                       |                       |                       |                    |

#### Beam Termination Point (select one)

| Hall A | Hall B | Hall C | BSY Dump | NE Stub Dump | 45 MeV Dump | Other (specify) |
|--------|--------|--------|----------|--------------|-------------|-----------------|
| X      |        |        |          |              |             |                 |

**Type of Test (select one)**

Invasive (disrupts beam delivery)

Non-invasive (does not disrupt beam delivery)

to A only

**Time Required**

few hours

**Preferred Time of Test**

Tuesday AM or owl Saturday

**Staff Required to Execute the Test (including contact info)**

Ops

**Controlled Access Requirements**

none

**Hardware and/or Software Changes Required**

**NOTE:** If software changes are part of the test plan, include the name of the application, the old revision level, the new revision level, and if applicable, whether or not it is possible to roll back to the old revision level (are there hardware limitations, etc.).

none

**Setup Procedure**

1. make sure you have a recent save of the hall A line

**Test Procedure**

1. Set MBC20H = <sup>1500</sup>~~760~~, MBT1H01 = <sup>-1300</sup>~~-680~~, MBC18V = <sup>5700</sup>~~3400~~, MBC20V = <sup>-5400</sup>~~-3000~~, MQA1H01 = MQM1H02 = MQO1H03 = MQO1H03A = 0.
2. Cycle these magnets, then take them off loop.
3. Put pulsed beam through. This should result in a beam <sup>3</sup>~~4~~ mm off axis in Y at 1H01 and subsequent BPM's. X position, since we don't have MBC1H18H, will be about 4mm off axis and brought back with 1H01 corrector. Take a screen shot of the BPM pattern.
4. Try to adjust the four quads, one at a time using 50 G-cm steps, to bring the beam to <sup>in a constant</sup>~~a constant~~ value in all BPM's beginning with 1H02. Don't move any quad more than 500 G-cm, and set each back to zero before moving on to the next. Take screen shots after optimizing each quad. Elog the quad values needed to achieve beam parallel to the axis.
5. Set the first five magnets listed in step 1 (the standard trims) to zero and cycle.
6. Set 1C20H = <sup>500</sup>~~500~~. Set MQA1H01 = <sup>10200</sup>~~4170~~. The beam should be at zero at

3400 to 1000

will

1H03B BPM. If it isn't, adjust MQA1H01 until it is, and elog the value. Then zero 1C20H and MQA1H01, including cycling.

7. set 1C20V = <sup>1200</sup>500. Set MQM1H02 = <sup>-10990</sup>-4505. Adjust the quad (off loop) until the beam is at zero at IPM 1H03B. Elog the value needed. Zero the quad. If time permits, cycle. If not, just set to zero off loop. Elog the BPM screen to show effect of residual field.
8. set MQO1H03 = <sup>-11340</sup>-4565. Adjust the quad (off loop) until the beam is at zero at IPM 1H03B. Elog the value. Zero the quad. If time permits, cycle. If not, just set to zero off loop. Elog the BPM screen to show effect of residual field.
9. set 1C20V = 0 and 1C20H = <sup>1200</sup>500. Cycle them.
10. set MQO1H03A = <sup>11238</sup>4600. Adjust the quad (off loop) until the beam is at zero at IPM 1H03B. Elog the value. Zero the quad. If time permits, cycle. If not, just set to zero off loop. Elog the BPM screen to show effect of residual field.
11. restore the end of the hall A line using the save set noted in the Setup section. Cycle the eight magnets used in this test.

### Backout Procedure

1. see step 11 above.

### Test Results

1) Mueller feed mag has no test  
 2) had shows can vary 5-8% from  
 current set point  
 3) need BPP sensors.  
 4) took test with zeros 1H03 BPM/sets  
 but tried to depress

11:38:36 April 03, 2000

# Hall A Rel. Pos. Spikes

(screen generated by c-code,  
Ilapps BPM team, Dec 21, 1999)

SEE Pass

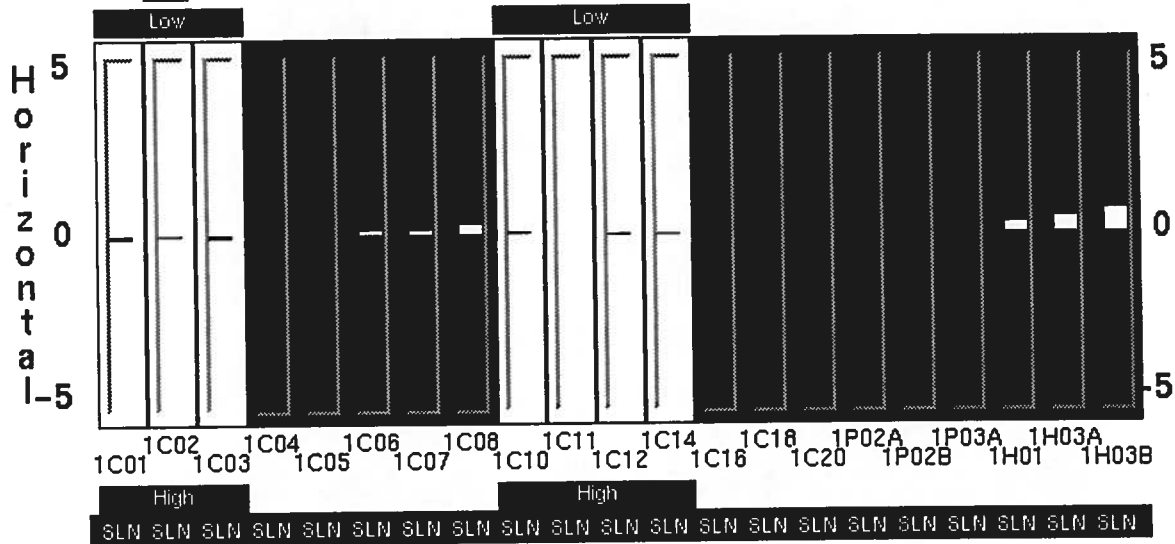
4.2 us, Pass 1

03Apr00 11:38:4

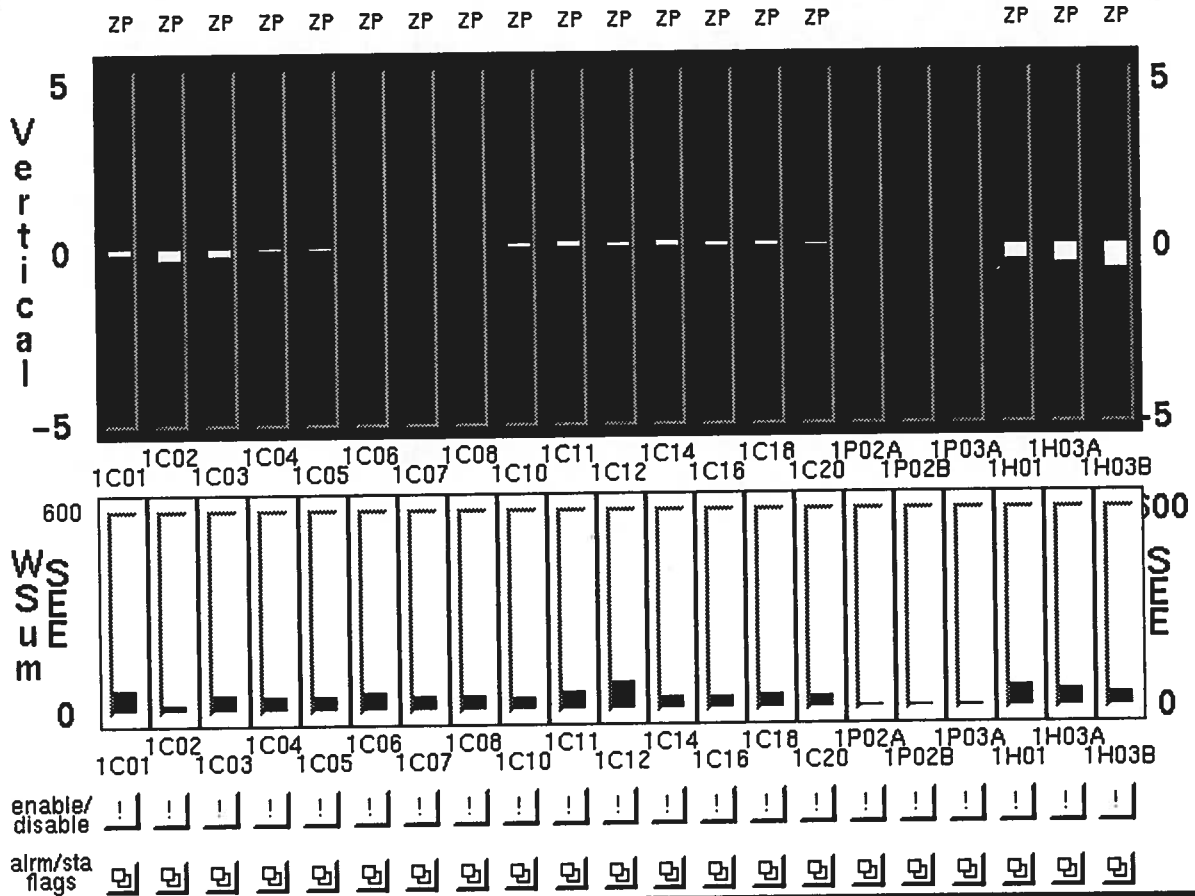
Utilities ☐

Zero Pos ☐

Back ☐



NP NP NP



Hall A Moller Polarimeter Target Location  
OUT

11:15:17 April 03, 2000

# Hall A Abs. Pos. Spikes

SEE Pass 4.2 us, Pass 1

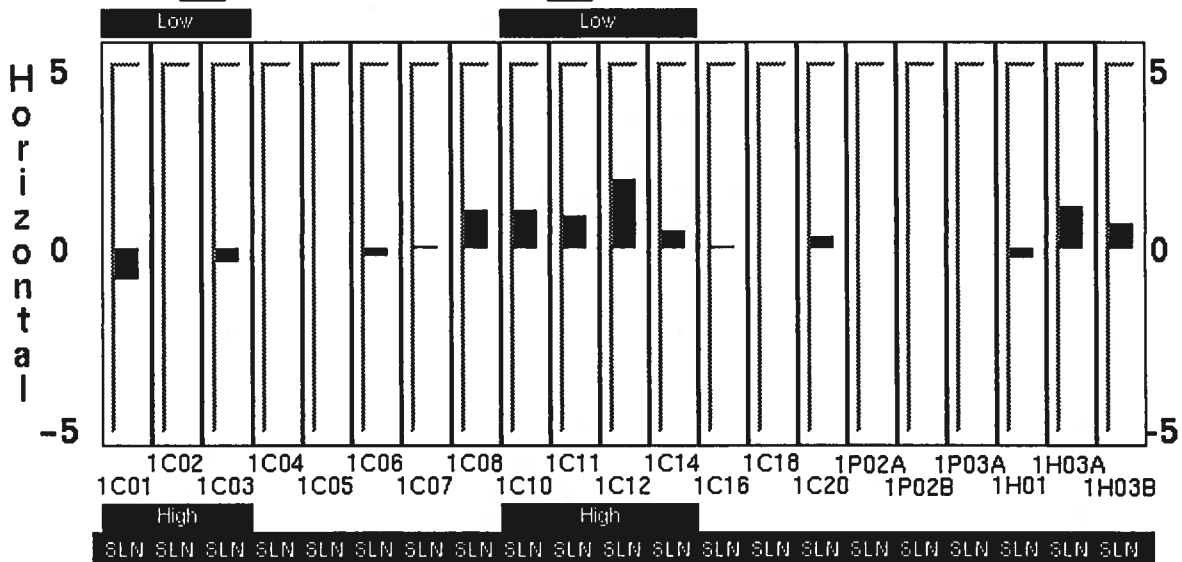
(screen generated by c-code,  
Ilapps BPM team, Dec 16, 1999)

03Apr00 11:15:2

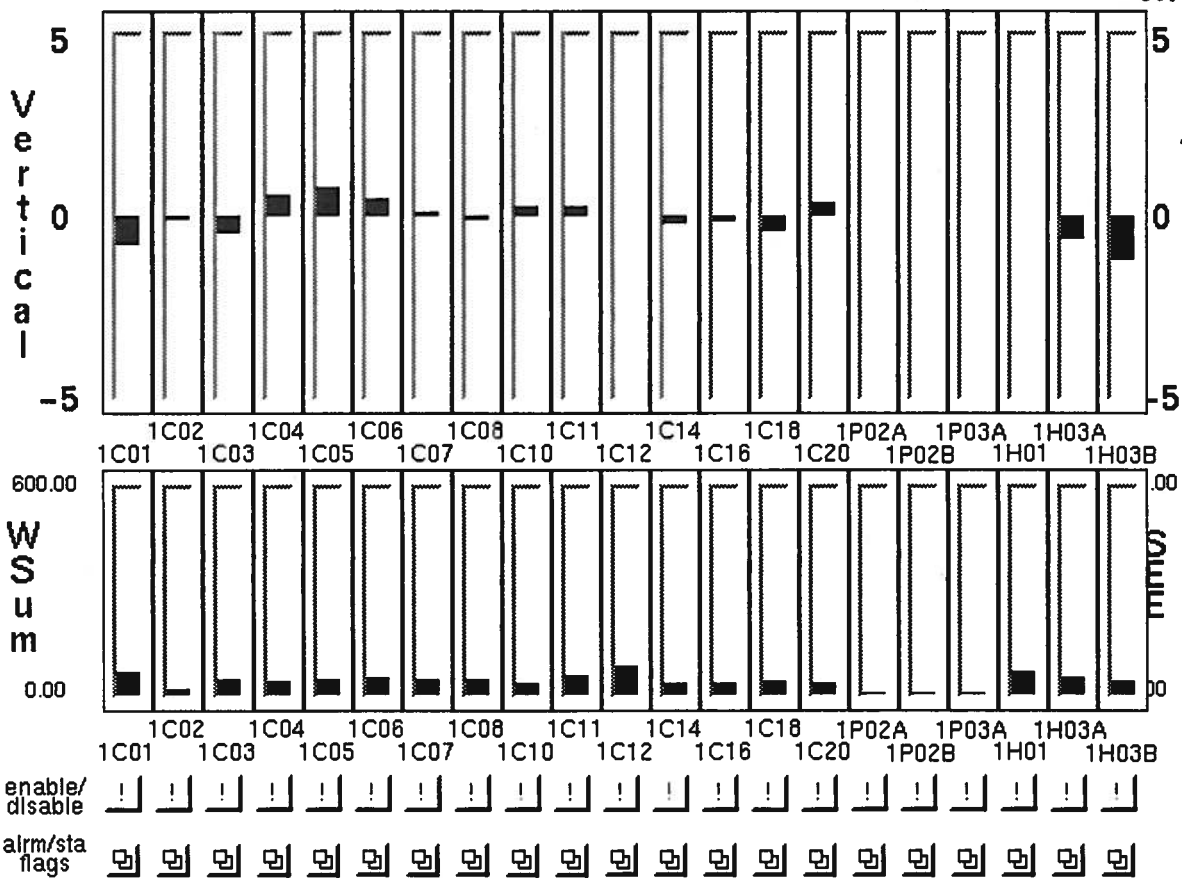
Utilities



Back



NP NP NP



Hall A Moller Polarimeter Target Location  
OUT

can not to  
Flatten, peaks  
zero

11:40:06 April 03, 2000

# Hall A Rel. Pos. Spikes

(screen generated by c-code,  
Ilappa BPM team, Dec 21, 1999)

SEE Pass

4.2 us, Pass 1

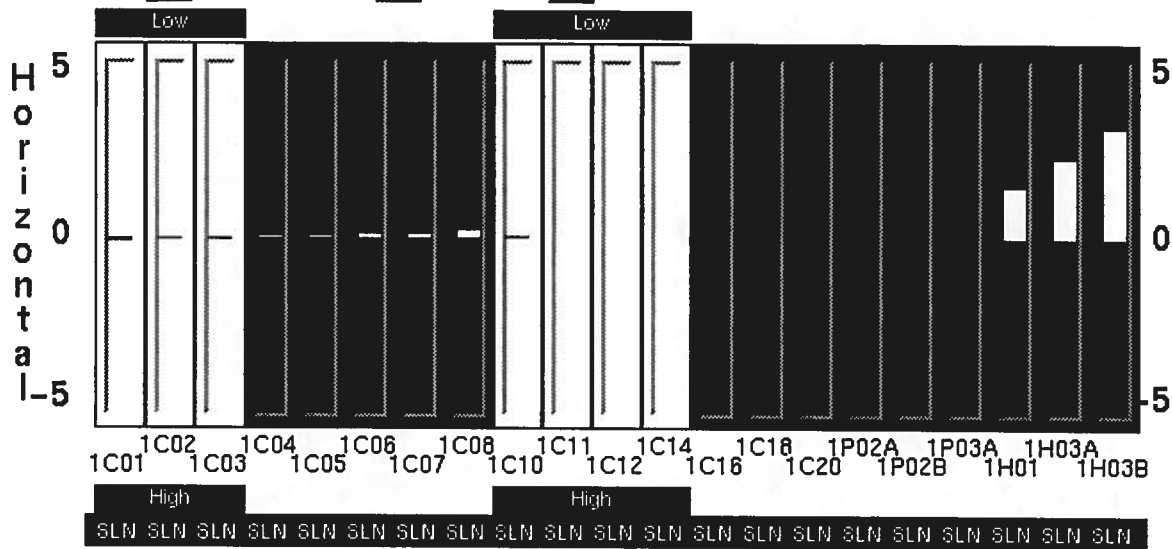
03Apr00 11:40:0

Utilities ☐

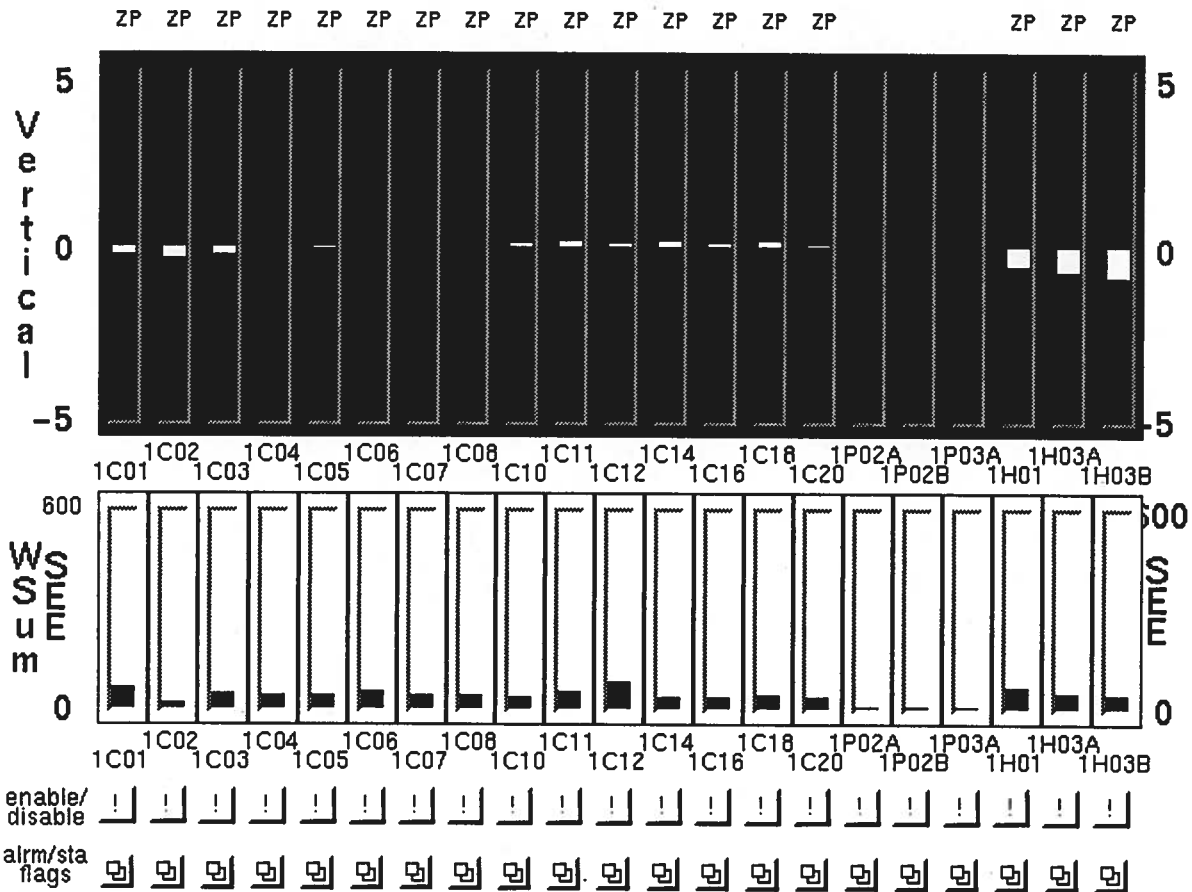
Zero Pos ☐

Back ☐

1C 20V -10



NP NP NP



Hall A Moller Polarimeter Target Location  
OUT

11:37:58 April 03, 2000

# Hall A Rel. Pos. Spikes

(screen generated by c-code,  
Ilapps BPM team, Dec 21, 1999)

SEE Pass

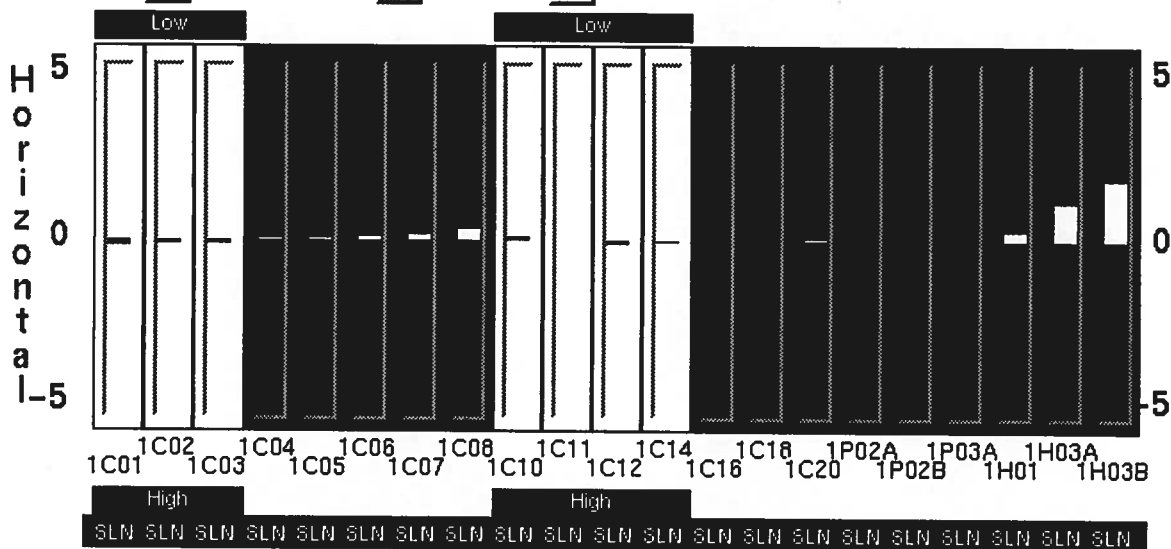
4.2 us, Pass 1

03Apr00 11:37:5

Utilities ☐

Zero Pos ☐

Back ☐

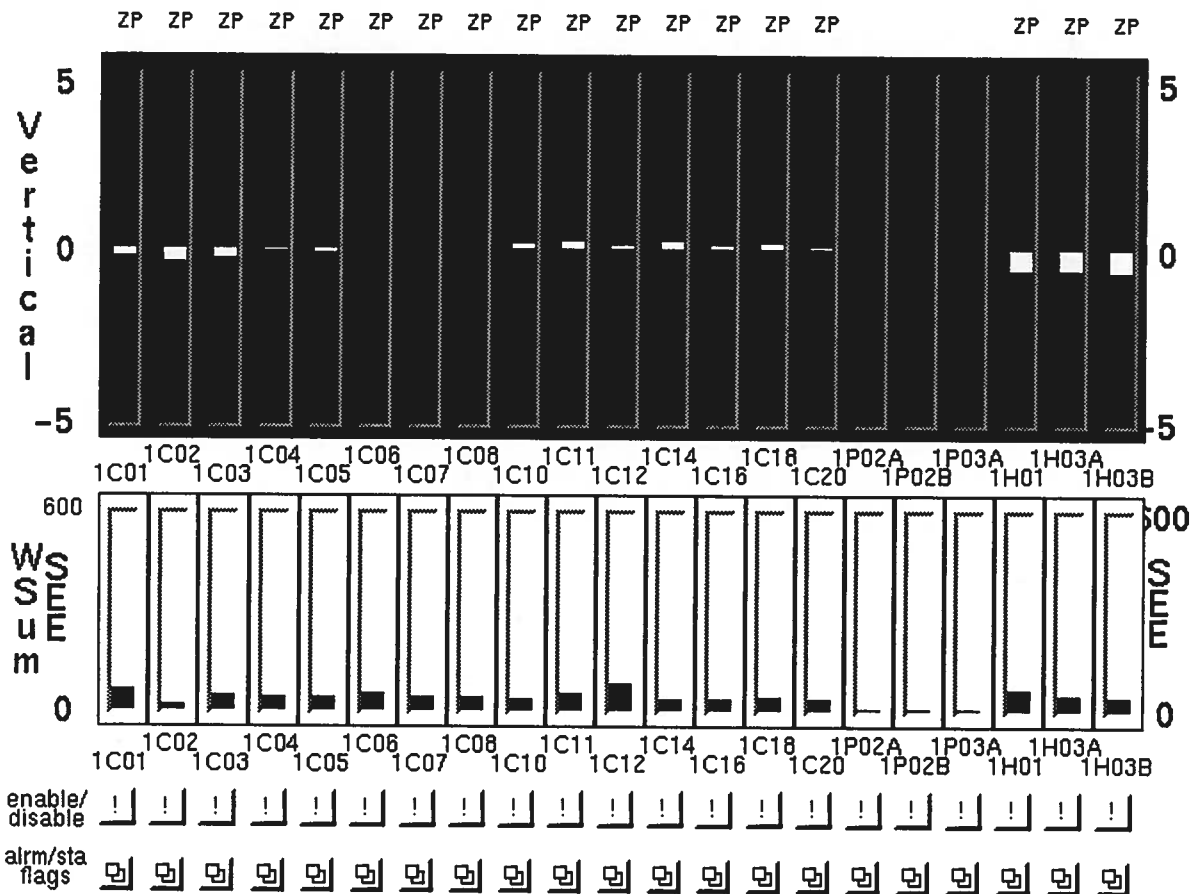


1H01/02/03

1C20H-90C

(Flat)

NP NP NP



Hall A Moller Polarimeter Target Location  
OUT

11:18:29 April 03, 2000

# Hall A Rel. Pos. Spikes

(screen generated by c-code,  
Ilapps BPM team, Dec 21, 1999)

SEE Pass

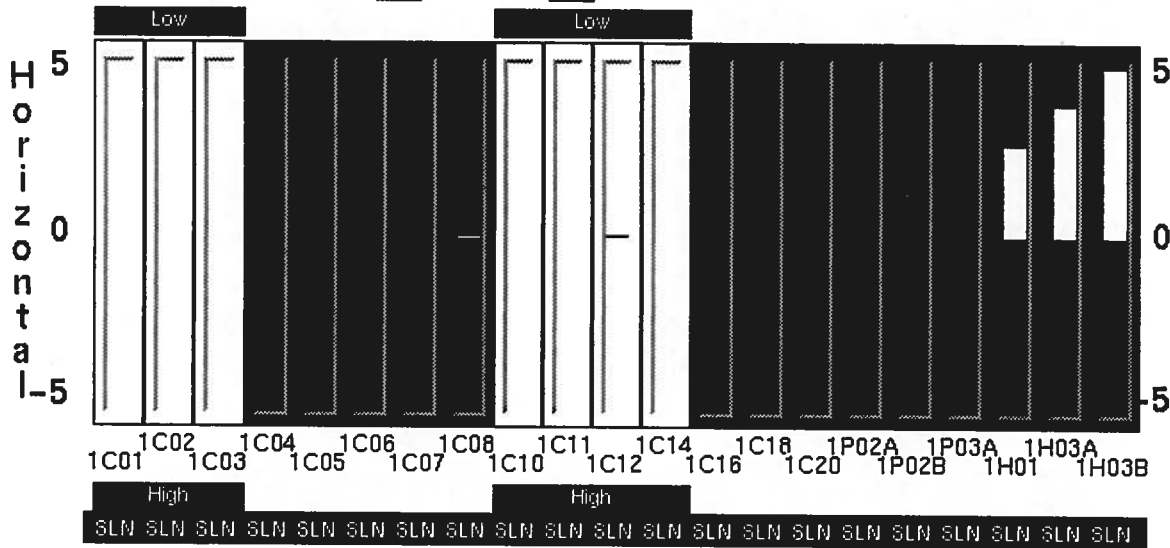
4.2 us, Pass 1

03Apr00 11:18:3

Utilities ☐

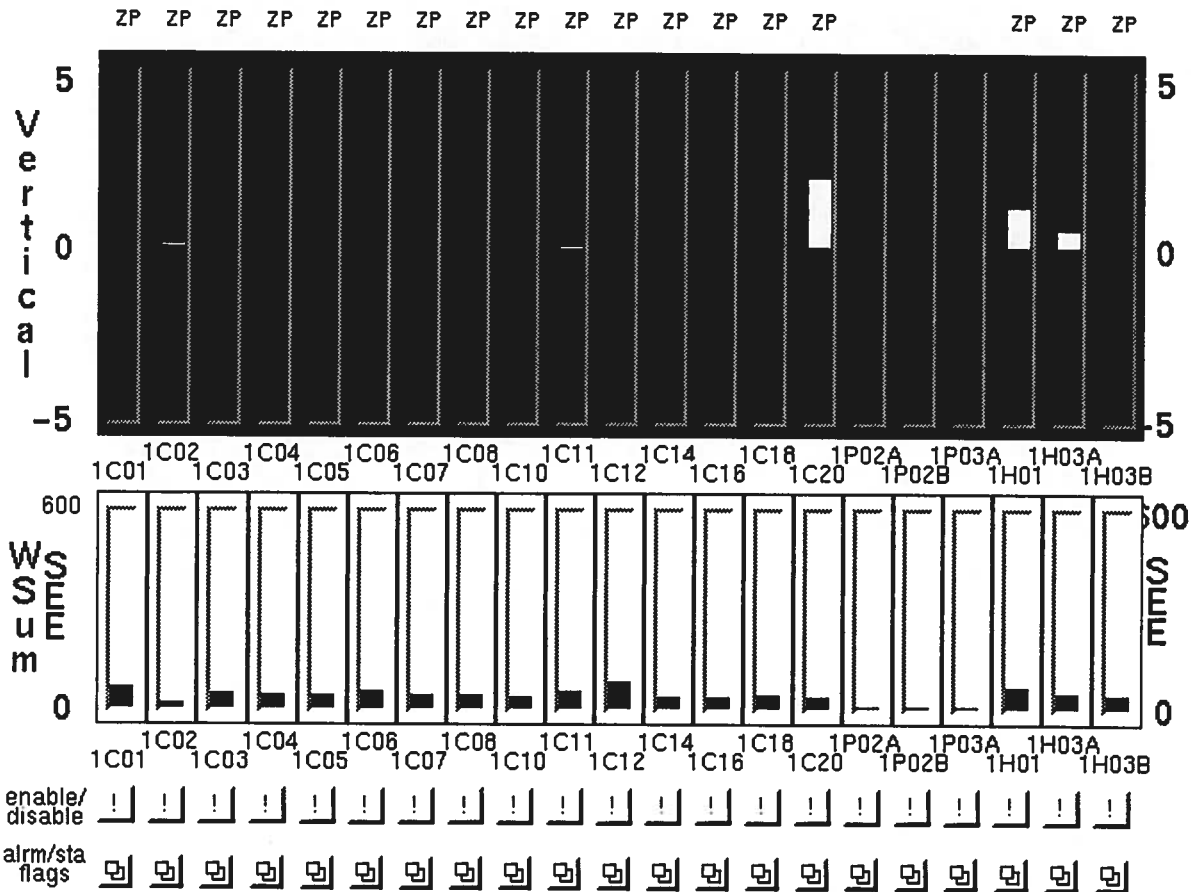
Zero Pos ☐

Back ☐



NP NP NP

*can't give  
11 beam*



Hall A Moller Polarimeter Target Location

OUT

11:12:21 April 03, 2000

# Hall A Abs. Pos. Spikes

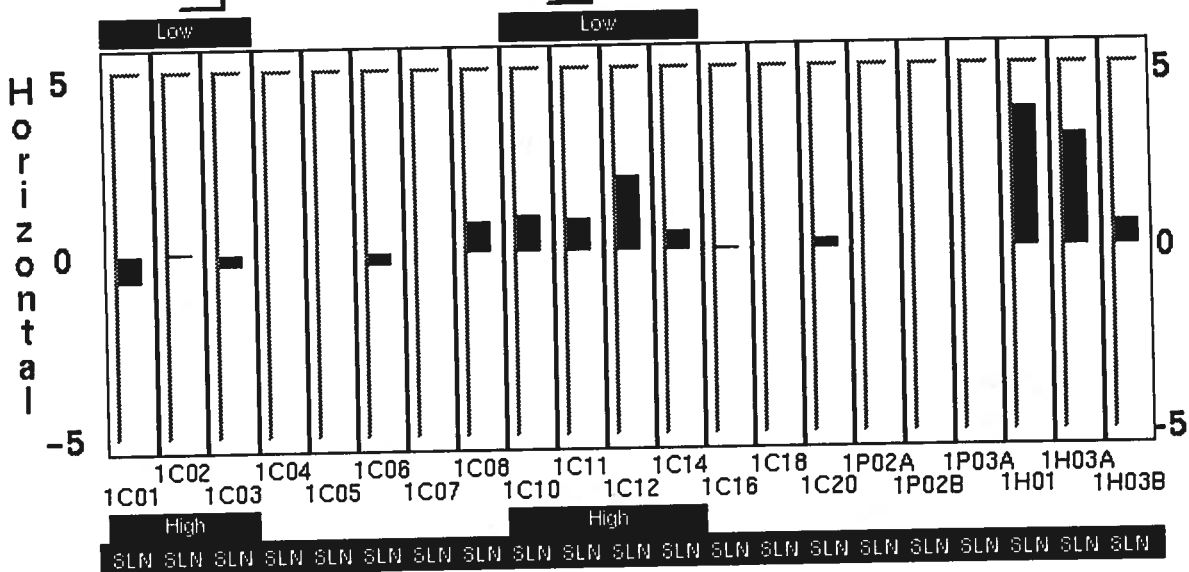
(screen generated by c-code,  
lapps BPM team, Dec 16, 1999)

SEE Pass 4.2 us, Pass 1

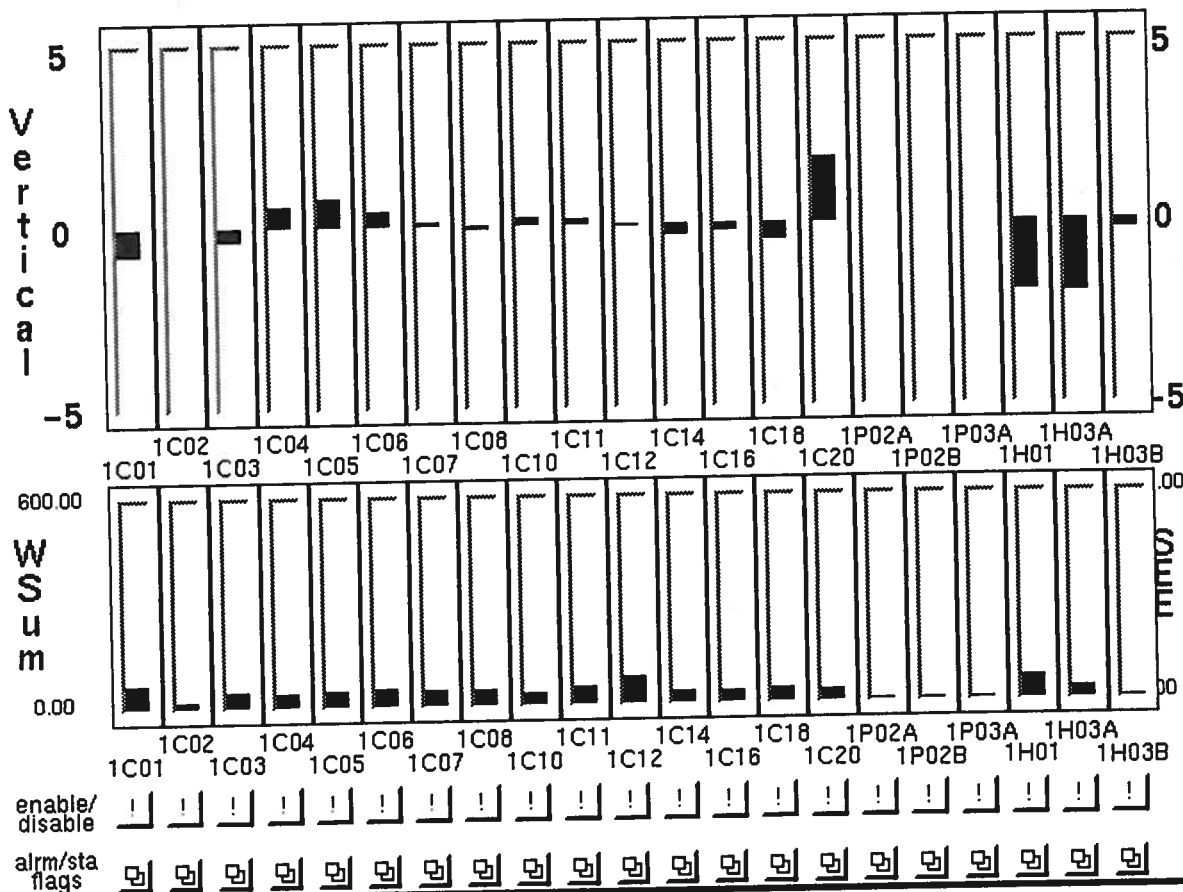
03Apr00 11:12:2

Utilities

Back



NP NP NP



Hall A Moller Polarimeter Target Location  
OUT